

MAT-MEK4270 Oblig 1

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1.2.3

$$\begin{aligned}u &= e^{i(k_x x + k_y y - \omega t)} \\u_{tt} &= -\omega^2 e^{i(k_x x + k_y y - \omega t)} \\&= -c^2(k_x^2 + k_y^2) e^{i(k_x x + k_y y - \omega t)} \\&= -c^2 k_x^2 e^{i(k_x x + k_y y - \omega t)} - c^2 k_y^2 e^{i(k_x x + k_y y - \omega t)} \\&= c^2(u_{xx} + u_{yy})\end{aligned}$$

1.2.4

$$u_{ij} = e^{i(kh(i+j) - \tilde{\omega} n \Delta t)}$$

Jeg setter opp den numeriske versjonen av likningen

$$\begin{aligned}&c^2 \left(\frac{e^{i(kh(i+1+j) - \tilde{\omega} n \Delta t)} - 2e^{i(kh(i+j) - \tilde{\omega} n \Delta t)} + e^{i(kh(i-1+j) - \tilde{\omega} n \Delta t)}}{\Delta x^2} + \right. \\&\quad \left. \frac{e^{i(kh(i+j+1) - \tilde{\omega} n \Delta t)} - 2e^{i(kh(i+j) - \tilde{\omega} n \Delta t)} + e^{i(kh(i+j-1) - \tilde{\omega} n \Delta t)}}{\Delta y^2} \right) \\&= \frac{e^{i(kh(i+j) - \tilde{\omega}(n+1)\Delta t)} - 2e^{i(kh(i+j) - \tilde{\omega} n \Delta t)} + e^{i(kh(i+j) - \tilde{\omega}(n-1)\Delta t)}}{\Delta t^2}\end{aligned}$$

Jeg setter inn for $\Delta x = \Delta y$:

$$\begin{aligned}&\frac{2e^{i(kh(i+1+j) - \tilde{\omega} n \Delta t)} - 4e^{i(kh(i+j) - \tilde{\omega} n \Delta t)} + 2e^{i(kh(i-1+j) - \tilde{\omega} n \Delta t)}}{\Delta x^2} \\&= \frac{e^{i(kh(i+j) - \tilde{\omega}(n+1)\Delta t)} - 2e^{i(kh(i+j) - \tilde{\omega} n \Delta t)} + e^{i(kh(i+j) - \tilde{\omega}(n-1)\Delta t)}}{\Delta t^2}\end{aligned}$$

Dermed:

$$\begin{aligned}c^2 \frac{2e^{ikh} - 4 + 2e^{-ikh}}{\Delta x^2} &= \frac{e^{-i\tilde{\omega}\Delta t-2} - 2 + e^{i\tilde{\omega}\Delta t}}{\Delta t^2} \\CFL^2(2e^{ikh} - 4 + 2e^{-ikh}) &= e^{-i\tilde{\omega}\Delta t-2} - 2 + e^{i\tilde{\omega}\Delta t}\end{aligned}$$

Jeg bruker at $\text{CFL} = \frac{1}{\sqrt{2}}$:

$$e^{ikh} - 2 + e^{-ikh} = e^{-i\tilde{\omega}\Delta t} - 2 + e^{i\tilde{\omega}\Delta t}$$

Dermed:

$$kh = \tilde{\omega}\Delta t$$

$$\tilde{\omega} = \frac{h}{\Delta t}k$$

Jeg bruker definisjonen av CFL:

$$\begin{aligned}\text{CFL} &= c \frac{\Delta t}{h} \\ \frac{1}{\sqrt{2}} &= c \frac{\Delta t}{h} \\ h &= \sqrt{2}c\Delta t\end{aligned}$$

Jeg setter dette inn for h :

$$\tilde{\omega} = \frac{\sqrt{2}c\Delta t}{\Delta t}k = \sqrt{2}ck$$

Jeg bruker at $k_x = k_y$:

$$\tilde{\omega}^2 = 2c^2k^2 = c^2(k_x^2 + k_y^2) = \omega$$